MTH621

—/—/202 : Q1:

Q1: $f(x) = \begin{cases} 0 & \text{if } x \text{ is irrational} \\ 1 & \text{if } x \text{ is rational} \end{cases}$

Find $\int_a^b f(x)dx$

Sol: Let $P = \{x_0, x_1, \dots x_n\}$ be a partition of $[a, b]$. Since every interval contains both rational and irrational numbers.

$$m_j = 0, \quad M_j = 1, \quad 1 \le j \le n.$$

$$S(P) = \sum_{j=1}^{n} 1\cdot(x_j - x_{j-1}) = b-a$$

$$s(P) = \sum_{j=1}^{n} 0\cdot(x_j - x_{j-1}) = 0$$

all upper sums equal $b-a$ and all lower sums equal 0,

$$\overline{\int_a^b} f(x)dx = b-a$$

$$\underline{\int_a^b} f(x)dx = 0$$

Q2

The Function

$$g(x) = \begin{cases} x+1 & , x>0 \\ x-1 & , x<0 \end{cases}$$

Sol: Every open interval containing $x_0 = 0$ also contains point x_1 and x_2 such that $|g(x_1) - g(x_2)|$ is as close 2, we as please, therefore $\lim_{x \to x_0} g(x)$ does not exist.

—/—/202 3 Day:

Calculate the derivate of
$h(x) = \sin\frac{1}{x}, \quad x \neq 0$

Sol: $f(x) = \sin x$ and $g(x) = \frac{1}{x}, x \neq 0$

$h(x) = f(g(x)) = \sin\frac{1}{x}, x \neq 0$

$h'(x) = f'(g(x))\,g'(x) = \left(\cos\frac{1}{x}\right)\left(\frac{-1}{x^2}\right), x \neq 0$

4

Theorem

If f is differentiable at x_0, then f is continous at x_0.

Sol: proof: $f(x) = f(x_0) + [f'(x_0) + E(x)](x - x_0)$,

where E is defind on a neighborhood of x_0

$\lim_{x \to x_0} E(x) = E(x_0) = 0$

where

$$E(x) = \begin{cases} \frac{f(x) - f(x_0)}{x - x_0} - f'(x_0), & x \in D_f,\ x \neq x_0 \\ 0 & x = x_0 \end{cases}$$

5

Monotonic Function

A function f is nondecreasing on an interval I if

$f(x_1) \leq f(x_2)$ whenever $x_1, x_2 \in I$, $x_1 \leq x_2$,

On nonincreasing on I if

$f(x_1) \geq f(x_2)$ wherever x_1 and x_2 in I $x_1 \leq x_2$.

6

Ex# $f(x) = \begin{cases} x, & 0 \leq x \leq 1 \\ x+1 & 1 \leq x \leq 2. \end{cases}$

Sol then function

$F(x) = \int_0^x f(t)\,dt = \begin{cases} x^2/2 & 0 \leq x \leq 1 \\ x^2/2 + x - 1 & 1 \leq x \leq 2. \end{cases}$

is continuous on $[0,2]$.

$F'(x) = \begin{cases} x = f(x) & 0 \leq x \leq 1 \\ x+1 = f(x) & 1 \leq x \leq 2 \end{cases}$

$F'_+(0) = \lim_{x \to 0^+} \frac{F(x) - F(0)}{x}$

$\lim_{x \to 0^+} \frac{(x^2/2) - 0}{x} = 0, \quad f(0)$

$F'_-(2) = \lim_{x \to 2^-} \frac{F(x) - F(2)}{x}$

$= \lim_{x \to 2^-} \frac{(x^2/2) + x - 1 - 3}{x - 2}$

$= \lim_{x \to 2^-} \frac{x+4}{2}$

$= 3$

$f(2)$

F does not have a derivate at $x = 1$

f is continus, $F'_-(1) = 1$, $F'_+(1) = 2$.

—/—1202

(7)

Evalute the following Limit

$$\lim_{x\to\infty} x^{1/x}$$

Sol: $x^{1/x} = \exp\left(\frac{\log x}{x}\right)$

$$= \lim_{x\to\infty} \frac{\log x}{x}$$

$$= \lim_{x\to\infty} \frac{1/x}{1} = (0) \text{ Ans}$$

(8)

Evaluete the following Limit.

$$\lim_{x\to 1} x^{1(x-1)}$$

Sol: $x^{1/x-1} = \exp\left(\frac{\log x}{x-1}\right)$

$$\lim_{x\to 1} \frac{\log x}{x-1}$$

$$\lim_{x\to 1} \frac{1/x}{x-1} = 1$$

it follows that

$$\lim_{x\to 1} x^{1/(x-1)} = e^{1} = e$$

(9)

Q: Check the uniform continuity of the function.

Sol:

for the function $f(x)$, we have $|f(x) - f(x')| = 2|x - x'| < \varepsilon$

if $|x - x'| < \varepsilon/2$

the function $f(x)$ is uniformly continus on $(-\infty, \infty)$.

(10)

$f(x) = \log x$ and $g(x) = \frac{1}{1-x^2}$

Can we define $f \circ g$?

Sol: The domain of the given function we

$D_f = (0, \infty)$ and $D_g = x \in R / x \neq \pm 1$

Since $g(x) > 0$ if $x \in T = (-1, 1)$ the composite function $f \circ g$ is defind on $(-1, 1)$ by

$$(f \circ g)(x) = \log x \frac{1}{1-x^2}$$

$g \circ f = ?$

the function $g \circ f$ on $(0, 1/e) \cup (1/e, e) \cup (e, \infty)$ by

$$(g \circ f)(x) = \frac{1}{1-(\log x)^2}.$$

—/—/1202

(11)

Ex #

Show that the function $g(x) = x^2$ is uniformly continuous on $[-r, r]$ for finite r.

Sol:

$$|g(x) - g(x')| = |x^2 - (x')^2|$$

$$|x - x'||x + x'| \le 2r|x - x'|$$

So,

$$|g(x) - g(x')| < \varepsilon \text{ if}$$

$$|x - x'| < \delta = \frac{\varepsilon}{2r}$$

and $-r \le x, x' \le r$.

(12)

For the function

$$f(x) = \sqrt{x}, \quad g(x) = \frac{9 - x^2}{x + 1}$$

check the continuity fog?

Sol:

The function f is continuous for $x > 0$ and the function g is continuous for $x \ne -1$.

Since $g(x) > 0$ if $x < -3$ on $-1 < x < 3$. Composite function

$$f \circ g(x) = \sqrt{\frac{9 - x^2}{x + 1}}$$

is continuous on $(-\infty, -3) \cup (-1, 3)$

(13) Day:

The function $g(x) = x^2$ is not uniformly continuous on $(-\infty, \infty)$.
if $\delta > 0$ there are real numbers x and x' such that
$|x - x'| = \delta/2$ and $|g(x) - g(x')| \geq 1$.
to this end,
$|g(x) - g(x')| = |x^2 - (x')^2| = |x - x'|(x + x')$
if $|x - x'| = \delta/2$, $x, x' > 1/\delta$

$|x - x'||x + x'| > \delta/2 \left(\frac{1}{\delta} + \frac{1}{\delta}\right) = 1$

(14)

Find the derivate of the line
$y = mx + c$.

Sol:
$$= \frac{f(x) - f(x_0)}{x - x_0}$$

$$= \frac{mx + c - (mx_0 + c)}{x - x_0}$$

$$\frac{m(x - x_0)}{x - x_0} = m$$

$$f'(x_0) = \lim_{x \to x_0} m = m$$

(15)

if n is a positive integer find the derivate of the function,
$$f(x) = x^n$$

___/___/202

Day: ______

$$\frac{f(x)-f(x_0)}{x-x_0} = \frac{x^n - x_0^n}{x-x_0} = \frac{x-x_0}{x-x_0}\sum_{k=0}^{n-1} x^{n-k-1} x_0^k,$$

$$f'(x_0) = \lim_{x\to x_0} \sum_{k=0}^{n-1} x^{n-k-1} x_0^k$$

$$= n x_0^{n-1}$$

this hold every x_0,

$f'(x) = xn^{n-1}$ or $\frac{d}{dx}(x^n)$

$$\boxed{n x^{n-1}}$$

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Evaluate $\lim_{x\to 0} \frac{4-4\cos x - 2\sin^2 x}{x^4}$

Sol:

$$\lim_{x\to 0} \frac{4\sin x - 4\sin x\cos x}{4x^3}$$

$$= \left(\lim_{x\to 0}\frac{\sin x}{x}\right)\left(\lim_{x\to 0}\frac{1-\cos x}{x^2}\right)$$

$$= \left(\lim_{x\to 0}\frac{\sin x}{x}\right)\left(\lim_{x\to 0}\frac{\sin x}{2x}\right)$$

$$\frac{1}{2}\left(\lim_{x\to 0}\frac{\sin x}{x}\right)^2$$

$$\frac{1}{2}(1)^2 = \frac{1}{2}$$

Riemann Integration

5.1 Riemann Sums

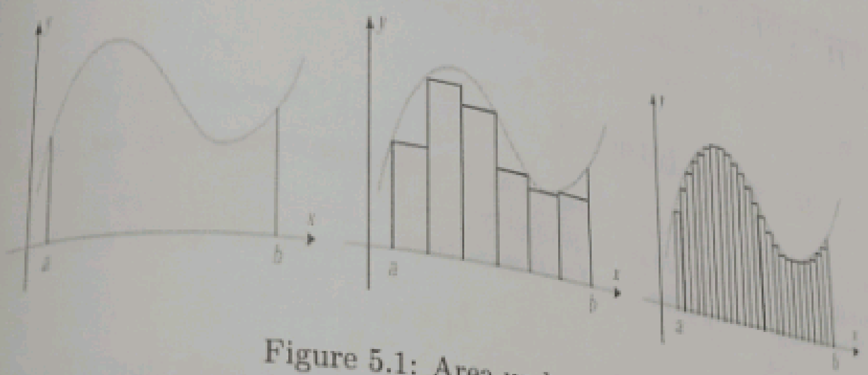

Figure 5.1: Area under the curve

Partition of an interval: Let f be a real valued function defined on a finite interval $[a, b]$.

A partition of $[a, b]$ is a set of subintervals

$$[x_0, x_1],\ [x_1, x_2], \ldots, [x_{n-1}, x_n],$$

where

$$a = x_0 < x_1 \cdots < x_n = b.$$

Thus, any set of $n + 1$ points satisfying $a = x_0 < x_1 \cdots < x_n = b$. defines a partition P of $[a, b]$, which we denote by

$$P = \{x_0, x_1, \ldots, x_n\}.$$

The points x_0, x_1, $\ldots$, x_n are the *partition points* of P.

The largest of the lengths of the subintervals $[x_0, x_1]$, $[x_1, x_2], \ldots, [x_{n-1}, x_n]$, is the *norm* of P, written as $\|P\|$; thus,

$$\|P\| = \max_{1 \le i \le n} (x_i - x_i - 1).$$

Quiz

Refinement of a partition: If P and P' are partitions of $[a, b]$, then P' is a *refinement of* P if every partition point of P is also a partition point of P'; that is, if P' is obtained by inserting additional points between those of P.

Defi Riemann sums: If f is defined on $[a, b]$, then a sum

$$\sigma = \sum_{j=1}^{n} f(c_j)(x_j - x_{j-1}),$$

where

$$x_{j-1} \leq c_j \leq x_j, \quad 1 \leq j \leq n,$$

is a *Riemann sum of f over the partition $P = \{x_0, x_1, \dots, x_n\}$*. (Occasionally we will say more simply that σ is a Riemann sum of f over $[a, b]$.)

Since c_j can be chosen arbitrarily in $[x_j, x_{j-1}]$, there are infinitely many Riemann sums for a given function f over a given partition P.

5.2 Riemann Integral

Let f be defined on $[a, b]$. We say that f is *Riemann integrable on* $[a, b]$ if there is a number L with the following property:

For every $\varepsilon > 0$, there is a $\delta > 0$ such that

$$|\sigma - L| < \varepsilon.$$

If σ is any Riemann sum of f over a partition P of $[a, b]$ such that $\|P\| < \delta$. In this case, we say that L is *the Riemann integral of f over* $[a, b]$, and write

$$\int_a^b f(x)\, dx = L.$$

Qu2 A0

Theorem: Prove that the Riemann integral $\int_a^b f(x)dx$, if it exists, is unique.

Proof: See Lecture.

- Prove: If $\int_a^b f(x)\, dx$ exists, then for every $\varepsilon > 0$, there is a $\delta > 0$ such that $|\sigma_1 - \sigma_2| < \varepsilon$ if σ_1 and σ_2 are Riemann sums of f over partitions P_1 and P_2 of $[a, b]$ with norms less than δ.

Example: Determine f is Riemann integrable over the given interval or not? Find $\int_a^b f(x)dx$, where

$$f(x) = 1, \quad a \leq x \leq b.$$

Solution: For the given function, we have

$$\sum_{j=1}^{n} f(c_j)(x_j - x_{j-1}) = \sum_{j=1}^{n} (x_j - x_{j-1}).$$

$$= -x_0 + (x_1 - x_1) + (x_2 - x_2) + \cdots + (x_{n-1} - x_{n-1}) + x_n$$
$$= x_n - x_0 = b - a.$$

Thus, every Riemann sum of f over any partition of $[a,b]$ equals $b-a$, so

$$\int_a^b dx = b - a.$$

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Example: For the function

$$f(x) = x, \quad a \le x \le b,$$

Riemann sums are of the form

$$\sigma = \sum_{j=1}^{n} c_j(x_j - x_{j-1}). \tag{5.1}$$

Since $x_{j-1} \le c_j \le x_j$ and $(x_j + x_{j-1})/2$ is the midpoint of $[x_{j-1}, x_j]$, we can write

$$c_j = \frac{x_j + x_{j-1}}{2} + d_j, \tag{5.2}$$

where

$$|d_j| \le \frac{x_j - x_{j-1}}{2} \le \frac{\|P\|}{2}. \tag{5.3}$$

Example: Substituting (5.2) into (5.1) yields

$$\begin{aligned} \sigma &= \textstyle\sum_{j=1}^{n} \frac{x_j + x_{j-1}}{2}(x_j - x_{j-1}) + \sum_{j=1}^{n} d_j(x_j - x_{j-1}) \\ &= \textstyle\frac{1}{2}\sum_{j=1}^{n}(x_j^2 - x_{j-1}^2) + \sum_{j=1}^{n} d_j(x_j - x_{j-1}). \end{aligned} \tag{5.4}$$

Because of cancelations, we have

$$\sum_{j=1}^{n}(x_j^2 - x_{j-1}^2) = b^2 - a^2,$$

so (5.4) can be rewritten as

Hence,

$$\left|\sigma - \frac{b^2-a^2}{2}\right| \leq \sum_{j=1}^{n} |d_j|(x_j - x_{j-1}) \leq \frac{\|P\|}{2}\sum_{j=1}^{n}(x_j - x_{j-1}) = \frac{\|P\|}{2}(b-a).$$

Therefore, every Riemann sum of f over a partition P of $[a,b]$ satisfies

$$\left|\sigma - \frac{b^2-a^2}{2}\right| < \varepsilon \quad \text{if} \quad \|P\| < \delta = \frac{2\varepsilon}{b-a}.$$

Hence,

$$\int_a^b x\,dx = \frac{b^2-a^2}{2}.$$

Quiz Qu Ans

Theorem: If f is unbounded on $[a,b]$, then f is not integrable on $[a,b]$.

Proof: We will show that if f is unbounded on $[a,b]$, P is any partition of $[a,b]$, and $M>0$, then there are Riemann sums σ and σ' of f over P such that

$$|\sigma - \sigma'| \geq M. \tag{5.5}$$

Let

$$\sigma = \sum_{j=1}^{n} f(c_j)(x_j - x_{j-1})$$

be a Riemann sum of f over a partition P of $[a,b]$.

There must be an integer i in $\{1,2,\dots,n\}$ such that

$$|f(c) - f(c_i)| \geq \frac{M}{x_i - x_{i-1}}$$

for some c in $[x_{i-1}x_i]$.

Because if there were not so, we would have

$$|f(x) - f(c_j)| < \frac{M}{x_j - x_{j-1}}, \quad x_{j-1} \leq x \leq x_j, \quad 1 \leq j \leq n.$$

Then

$$\begin{aligned} |f(x)| &= |f(c_j) + f(x) - f(c_j)| \leq |f(c_j)| + |f(x) - f(c_j)| \\ &\leq |f(c_j)| + \frac{M}{x_j - x_{j-1}}, \quad x_{j-1} \leq x \leq x_j, \quad 1 \leq j \leq n. \end{aligned}$$

which implies that

$$|f(x)| \leq \max_{1\leq j\leq n} |f(c_j)| + \frac{M}{x_j - x_{j-1}}, \quad a \leq x \leq b,$$

$$\sum_{j=1} f(c'_j)(x_j - x_{j-1})$$

over the same partition P, where

$$c'_j = \begin{cases} c_j, & j \neq i, \\ c, & j = i. \end{cases}$$

Since

$$|\sigma - \sigma'| = |f(c) - f(c_i)|(x_i - x_{i-1}),$$

hence due to

$$|f(c) - f(c_i)| \geq \frac{M}{x_i - x_{i-1}}$$

implies (5.5).

5.3 Upper and Lower Integrals

Upper and lower integrals: If f is bounded on $[a, b]$ and $P = \{x_0, x_1, \ldots, x_n\}$ is a partition of $[a, b]$, let

$$M_j = \sup_{x_{j-1} \leq x \leq x_j} f(x) \quad \text{and} \quad m_j = \inf_{x_{j-1} \leq x \leq x_j} f(x).$$

The *upper sum of f over P* is

Due

$$S(P) = \sum_{j=1}^{n} M_j(x_j - x_{j-1}),$$

and the *upper integral of f over*, $[a, b]$, denoted by

$$\overline{\int_a^b} f(x)\, dx,$$

is the infimum of all upper sums.

Define term lower integral? taken

Lower integrals: The *lower sum of f over P* is

$$s(P) = \sum_{j=1}^{n} m_j(x_j - x_{j-1}),$$

and the *lower integral of f over* $[a, b]$, denoted by

$$\underline{\int_a^b} f(x)\, dx,$$

is the supremum of all lower sums.

Remark: If $m \leq f(x) \leq M$ for all x in $[a, b]$, then

$$m(b-a) \leq s(P) \leq S(P) \leq M(b-a)$$

for every partition P. Thus, the set of upper sums of f over all partitions P of $[a, b]$ is bounded, as is the set of lower sums.

Therefore, definition of infimum and supremum imply that $\overline{\int_a^b} f(x)\,dx$ and $\underline{\int_a^b} f(x)\,dx$ exist, are unique, and satisfy the inequalities

$$m(b-a) \leq \overline{\int_a^b} f(x)\,dx \leq M(b-a)$$

and

$$m(b-a) \leq \underline{\int_a^b} f(x)\,dx \leq M(b-a).$$

Theorem: Let f be bounded on $[a, b]$, and let P be a partition of $[a, b]$. Then

Imp

1. The upper sum $S(P)$ of f over P is the supremum of the set of all Riemann sums of f over P.
2. The lower sum $s(P)$ of f over P is the infimum of the set of all Riemann sums of f over P.

Proof: If $P = \{x_0, x_1, \ldots, x_n\}$, then

$$S(P) = \sum_{j=1}^{n} M_j(x_j - x_{j-1}),$$

where

$$M_j = \sup_{x_{j-1} \leq x \leq x_j} f(x).$$

An arbitrary Riemann sum of f over P is of the form

$$\sigma = \sum_{j=1}^{n} f(c_j)(x_j - x_{j-1}),$$

where $x_{j-1} \leq c_j \leq x_j$.

Since $f(c_j) \leq M_j$, it follows that $\sigma \leq S(P)$. Now let $\varepsilon > 0$ and choose $\overline{c}_j$ in $[x_{j-1}, x_j]$ so that

$$f(\overline{c}_j) > M_j - \frac{\varepsilon}{n(x_j - x_{j-1})}, \quad 1 \leq j \leq n.$$

$$\begin{aligned} \sigma &= \sum_{j=1} f(c_j)(x_j - x_{j-1}) > \sum_{j=1} \left[M_j - \frac{\varepsilon}{n(x_j - x_{j-1})} \right] (x_j - x_{j-1}) \\ &= S(P) - \varepsilon. \end{aligned}$$

Now from definition $S(P)$ is the supremum of the set of Riemann sums of f over P.

P of $[a, b]$

$x)\,dx$ and

n

mann

sums

Example: Let

s.d

$$f(x) = \begin{cases} 0 & \text{if } x \text{ is irrational,} \\ 1 & \text{if } x \text{ is rational,} \end{cases}$$

Let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$. Since every interval contains both rational and irrational numbers

$$m_j = 0 \quad \text{and} \quad M_j = 1, \quad 1 \le j \le n.$$

Hence,

$$\begin{aligned} S(P) &= \sum_{j=1}^{n} 1 \cdot (x_j - x_{j-1}) = b - a \\ s(P) &= \sum_{j=1}^{n} 0 \cdot (x_j - x_{j-1}) = 0. \end{aligned}$$

Since all upper sums equal $b - a$ and all lower sums equal 0, from definition we have

$$\overline{\int_a^b} f(x)\,dx = b - a \quad \text{and} \quad \underline{\int_a^b} f(x)\,dx = 0.$$

Example: Let f be defined on $[1, 2]$ by $f(x) = 0$ if x is irrational and $f(p/q) = 1/q$ if p and q are positive integers with no common factors.

If $P = \{x_0, x_1, \dots, x_n\}$ is any partition of $[1, 2]$, then $m_j = 0$, $1 \le j \le n$, so $s(P) = 0$; hence,

$$\underline{\int_1^2} f(x)\,dx = 0. \tag{5.6}$$

We now show that

$$\overline{\int_1^2} f(x)\,dx = 0$$

also.

Therefore, $\int_a^b f(x)\,dx$ exists.

Since ε can be any positive number, this implies that

$$\overline{\int_a^b} f(x)\,dx = \underline{\int_a^b} f(x)\,dx.$$

For the converse of the theorem try yourself

Theorem: If f is bounded on $[a,b]$, then f is integrable on $[a,b]$ if and only if for each $\varepsilon > 0$ there is a partition P of $[a,b]$ for which

$$S(P) - s(P) < \varepsilon.$$

Theorem: If f is continuous on $[a,b]$, then f is integrable on $[a,b]$.

Proof: Let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a,b]$.

Since f is continuous on $[a,b]$, there are points c_j and c_j' in $[x_{j-1}, x_j]$ such that

$$f(c_j) = M_j = \sup_{x_{j-1} \le x \le x_j} f(x)$$

and

$$f(c_j') = m_j = \inf_{x_{j-1} \le x \le x_j} f(x).$$

Therefore,

$$S(P) - s(P) = \sum_{j=1}^{n} \left[f(c_j) - f(c_j')\right](x_j - x_{j-1}). \tag{5.23}$$

Since f is uniformly continuous on $[a,b]$, there is for each $\varepsilon > 0$ a $\delta > 0$ such that

$$|f(x') - f(x)| < \frac{\varepsilon}{b-a}.$$

If x and x' are in $[a,b]$ and $|x - x'| < \delta$. If $\|P\| < \delta$, then $|c_j - c_j'| < \delta$ and, from (5.23),

$$S(P) - s(P) < \frac{\varepsilon}{b-a}\sum_{j=1}^{n}(x_j - x_{j-1}) = \varepsilon.$$

Hence, f is integrable on $[a,b]$.

Theorem: If f ... f is integrable on $[a, b]$ if and only if for each $\varepsilon > 0$ there is a partition P of $[a, b]$ for which

$$S(P) - s(P) < \varepsilon.$$

Theorem: If f is monotonic on $[a, b]$, then f is integrable on $[a, b]$. U2

Proof: Let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$. Since f is nondecreasing,

$$\begin{aligned} f(x_j) &= M_j = \sup_{x_{j-1} \le x \le x_j} f(x) \\ f(x_{j-1}) &= m_j = \inf_{x_{j-1} \le x \le x_j} f(x). \end{aligned}$$

Hence,

$$S(P) - s(P) = \sum_{j=1}^{n} (f(x_j) - f(x_{j-1}))(x_j - x_{j-1}).$$

Since $0 < x_j - x_{j-1} \le \|P\|$ and $f(x_j) - f(x_{j-1}) \ge 0$,

$$\begin{aligned} S(P) - s(P) &\le \|P\| \sum_{j=1}^{n} (f(x_j) - f(x_{j-1})) \\ &= \|P\|(f(b) - f(a)). \end{aligned}$$

Therefore,

$$S(P) - s(P) < \varepsilon \quad \text{if} \quad \|P\|(f(b) - f(a)) < \varepsilon,$$

so f is integrable on $[a, b]$.

Theorem: If f and g are integrable on $[a, b]$, then so is $f + g$, and

$$\int_a^b (f + g)(x)\, dx = \int_a^b f(x)\, dx + \int_a^b g(x)\, dx.$$

Proof: Any Riemann sum of $f + g$ over a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$ can be written as

$$\begin{aligned} \sigma_{f+g} &= \textstyle\sum_{j=1}^{n} [f(c_j) + g(c_j)](x_j - x_{j-1}) \\ &= \textstyle\sum_{j=1}^{n} f(c_j)(x_j - x_{j-1}) + \sum_{j=1}^{n} g(c_j)(x_j - x_{j-1}) \\ &= \sigma_f + \sigma_g, \end{aligned}$$

Definition of Riemann integ...
δ_1 and δ_2 such that

$$\left|\sigma_f - \int_a^b f(x)\,dx\right| < \frac{\varepsilon}{2} \quad \text{if} \quad \|P\| < \delta_1$$

and

$$\left|\sigma_g - \int_a^b g(x)\,dx\right| < \frac{\varepsilon}{2} \quad \text{if} \quad \|P\| < \delta_2.$$

If $\|P\| < \delta = \min(\delta_1, \delta_2)$, then

$$\begin{aligned}\left|\sigma_{f+g} - \int_a^b f(x)\,dx - \int_a^b g(x)\,dx\right| &= \left|\left(\sigma_f - \int_a^b f(x)\,dx\right) + \left(\sigma_g - \int_a^b g(x)\,dx\right)\right| \\ &\le \left|\sigma_f - \int_a^b f(x)\,dx\right| + \left|\sigma_g - \int_a^b g(x)\,dx\right| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,\end{aligned}$$

so the conclusion follows from Definition.

Example: Determine whether the function $f(x) = 1 + x$ is integrable on $[a, b]$?

Sol

Solution: If

$$f(x) = \sin x \quad \text{and} \quad g(x) = \frac{1}{x}, \quad x \neq 0,$$

then

$$h(x) = f(g(x)) = \sin\frac{1}{x}, \quad x \neq 0,$$

and

$$h'(x) = f'(g(x))g(x) = \left(\cos\frac{1}{x}\right)\left(-\frac{1}{x^2}\right), \quad x \neq 0.$$

Theorem: If f is integrable on $[a, b]$ and c is a constant, then cf is integrable on $[a, b]$ and

$$\int_a^b cf(x)\,dx = c\int_a^b f(x)\,dx.$$

Proof: See Lecture.

Theorem: If $f_1, f_2, \ldots, f_n$ are integrable on $[a, b]$ and $c_1, c_2, \ldots, c_n$ are constants, then $c_1f_1 + c_2f_2 + \cdots + c_nf_n$ is integrable on $[a, b]$ and

$$\begin{aligned}\int_a^b (c_1f_1 + c_2f_2 + \cdots + c_nf_n)(x)\,dx &= c_1\int_a^b f_1(x)\,dx \\ &\quad + \cdots + c_n\int_a^b f_n(x)\,dx.\end{aligned}$$

Example: If

$$f(x) = \begin{cases} x, & 0 \le x \le 1, \\ x+1, & 1 < x \le 2. \end{cases}$$

Then the function

$$F(x) = \int_0^x f(t)\,dt = \begin{cases} \frac{x^2}{2}, & 0 < x \le 1, \\ \frac{x^2}{2} + x - 1, & 1 < x \le 2, \end{cases}$$

is continuous on $[0, 2]$.

We can conclude

$$F'(x) = \begin{cases} x = f(x), & 0 < x < 1, \\ x + 1 = f(x), & 1 < x < 2. \end{cases}$$

$$\begin{aligned} F'_+(0) &= \lim_{x \to 0+} \frac{F(x) - F(0)}{x} = \lim_{x \to 0+} \frac{(x^2/2) - 0}{x} = 0 = f(0), \\ F'_-(2) &= \lim_{x \to 2-} \frac{F(x) - F(2)}{x - 2} = \lim_{x \to 2-} \frac{(x^2/2) + x - 1 - 3}{x - 2} \\ &= \lim_{x \to 2-} \frac{x + 4}{2} = 3 = f(2). \end{aligned}$$

F does not have a derivative at $x = 1$, where f is discontinuous, since

$$F'_-(1) = 1 \quad \text{and} \quad F'_+(1) = 2.$$

Theorem: Suppose that F is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b), and f is integrable on $[a, b]$.

Suppose also that

$$F'(x) = f(x), \quad a < x < b. \tag{5.38}$$

Then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

$0+$, $x \to b$, and $x \to \pm\infty$.

Example: Evaluate the following limit

$$\lim_{x \to 0+} x^x.$$

Solution: See Lecture.

Example: Evaluate the following limit

$$\lim_{x \to 1} x^{1/(x-1)}$$ Quiz

Solution:
Since

$$x^{1/(x-1)} = \exp\left(\frac{\log x}{x-1}\right)$$

and

$$\lim_{x \to 1} \frac{\log x}{x-1} = \lim_{x \to 1} \frac{1/x}{1} = 1,$$

it follows that

$$\lim_{x \to 1} x^{1/(x-1)} = e^1 = e.$$ Ans

Example: Evaluate the following limit

$$\lim_{x \to \infty} x^{1/x}.$$ Quiz

Solution:
Since

$$x^{1/x} = \exp\left(\frac{\log x}{x}\right)$$

and

$$\lim_{x \to \infty} \frac{\log x}{x} = \lim_{x \to \infty} \frac{1/x}{1} = 0,$$ Ans

$$f(x)g(x) = \frac{f(x)}{1/g(x)} \quad \text{or} \quad f(x)g(x) = \frac{g(x)}{1/f(x)},$$

since one of these ratios is of the form $0/0$ and the other is of the form ∞/∞ as $x \to b-$. Similar statements apply to limits as $x \to b+$, $x \to b$, and $x \to \pm\infty$.

S.Q

Example: Evaluate the limit

Quiz

$$\lim_{x\to 0+} x \log x.$$

Solution: The product $x \log x$ is of the form $0 \cdot \infty$ as $x \to 0+$. Converting it to an ∞/∞ form yields

$$\begin{aligned}
\lim_{x\to 0+} x \log x &= \lim_{x\to 0+} \frac{\log x}{1/x} \\
&= \lim_{x\to 0+} \frac{1/x}{-1/x^2} \\
&= -\lim_{x\to 0+} x = 0.
\end{aligned}$$

Ans

L.Q

Example: Evaluate the limit

$$\lim_{x\to\infty} x \log 1 + 1/x.$$

Solution: Converting it to a $0/0$ form yields

$$\begin{aligned}
\lim_{x\to\infty} x \log(1 + 1/x) &= \lim_{x\to\infty} \frac{\log(1+1/x)}{1/x} \\
&= \lim_{x\to\infty} \frac{[1/(1+1/x)]\,(-1/x^2)}{-1/x^2} \\
&= \lim_{x\to\infty} \frac{1}{1+1/x} = 1.
\end{aligned}$$

In this case, converting to an ∞/∞ form complicates the problem:

$$\begin{aligned}
\lim_{x\to\infty} x \log(1 + 1/x) &= \lim_{x\to\infty} \frac{x}{1/\log(1+1/x)} \\
&= \lim_{x\to\infty} \frac{1}{\left(\frac{-1}{[\log(1+1/x)]^2}\right)\left(\frac{-1/x^2}{1+1/x}\right)} \\
&= \lim_{x\to\infty} x(x+1)[\log(1+1/x)]^2 = ?
\end{aligned}$$

$$\lim_{x\to\infty} \frac{e^x}{x^2} = \infty.$$

...erally,

$$\lim_{x\to\infty} \frac{e^x}{x^\alpha} = \infty$$

...real number α

Quiz

...ple: Evaluate $\lim_{x\to 0} \frac{4-4\cos x - 2\sin^2 x}{x^4}$. Sol

...ion:

$$\begin{aligned}
\lim_{x\to 0} \frac{4-4\cos x-2\sin^2 x}{x^4} &= \lim_{x\to 0} \frac{4\sin x - 4\sin x\cos x}{4x^3} \\
&= \left(\lim_{x\to 0}\frac{\sin x}{x}\right)\left(\lim_{x\to 0}\frac{1-\cos x}{x^2}\right) \\
1 \quad &= \left(\lim_{x\to 0}\frac{\sin x}{x}\right)\left(\lim_{x\to 0}\frac{\sin x}{2x}\right) \\
&= \frac{1}{2}\left(\lim_{x\to 0}\frac{\sin x}{x}\right)^2 \\
&= \frac{1}{2}(1)^2 = \frac{1}{2}. \text{ Ans}
\end{aligned}$$

...that it is

...0, and

...ple: If

$$f(x) = x - x^2 \sin\frac{1}{x} \quad \text{and} \quad g(x) = \sin x,$$

$$f'(x) = 1 - 2x\sin\frac{1}{x} + \cos\frac{1}{x} \quad \text{and} \quad g'(x) = \cos x.$$

...efore, $\lim_{x\to 0} f'(x)/g'(x)$ does not exist. However,

$$\lim_{x\to 0}\frac{f(x)}{g(x)} = \lim_{x\to 0}\frac{1 - x\sin(1/x)}{(\sin x)/x} = \frac{1}{1} = 1.$$

4.9 Indeterminate Forms

Quiz

The quotient function f/g is *of the form* $0/0$ *as* $x \to b-$ if

$$\lim_{x\to b-} f(x) = \lim_{x\to b-} g(x) = 0,$$

Ay

or *of the form* ∞/∞ *as* $x \to b-$ if

$$\lim_{x\to b-} f(x) = \pm\infty$$

and

$$\lim_{x\to b-} g(x) = \pm\infty.$$

The corresponding definitions for $x \to b+$ and $x \to \pm\infty$ are similar.

If f/g is of one of these forms as $x \to b-$ and as $x \to b+$, then we say that it is of that form as $x \to b$.

Example: Evaluate the following limit

$$\lim_{x\to 0} \sin x/x.$$

Solution: The given limit limit $\lim_{x\to 0} \sin x/x$. is of the form $0/0$ as $x \to 0$, and L'Hospital's rule yields

$$\lim_{x\to 0} \frac{\sin x}{x} = \lim_{x\to 0} \frac{\cos x}{1} = 1.$$

Ay

Example: Evaluate the limit

$$\lim_{x\to -\infty} \frac{e^{-x}}{x}.$$

Solution: See Lecture.
Example: Evaluate the limit

$$\lim_{x\to +\infty} \frac{x^{-4/3}}{\sin(1/x)}.$$

Solution: See Lecture.

Example: Using L'Hospital's rule may lead to another indeterminate form; thus,

$$\lim_{x\to\infty} \frac{e^x}{x^2} = \lim_{x\to\infty} \frac{e^x}{2x}$$

4.8 Lipschitz Continuity

A function that satisfies inequality

$$|f(x) - f(x')| \leq M|x - x'|, \quad x, x' \in (a, b),$$

for all x and x' in an interval, where $M > 0$ is some real number is said to satisfy a *Lipschitz condition* on the interval.

Remark: A differentiable real valued function is Lipschitz continuous, indeed we have

differentiability at $x \Rightarrow$ Lipschits continuity at $x \Rightarrow$ Continuity at x

Lipschitz continuity: Continuity at $x \not\Rightarrow$ Lipschits continuity at $x \not\Rightarrow$ differentiability at x

Example: The function $f(x) = \sqrt{|x|}$ is continuous at $x = 0$, but not Lipschits continuous at $x = 0$.

The function $f(x) = |x|$ is Lipschits continuous at $x = 0$ but not differentiable at $x = 0$.

Theorem: If f and g are continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b), then

$$[g(b) - g(a)]f'(c)$$
$$= [f(b) - f(a)]g'(c)$$

for some c in (a, b).

Theorem: Suppose that f and g are differentiable and g' has no zeros on (a, b). Let

$$\lim_{x \to b-} f(x) = \lim_{x \to b-} g(x) = 0$$

or

$$\lim_{x \to b-} f(x) = \pm\infty \quad \text{and} \quad \lim_{x \to b-} g(x) = \pm\infty,$$

and suppose that

$$\lim_{x \to b-} \frac{f'(x)}{g'(x)} = L \quad (\text{finite or} \pm \infty).$$

Then

$$\lim_{x \to b-} \frac{f(x)}{g(x)} = L.$$

4.5 Rolle's Theorem Quiz

Theorem: Suppose that f is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b), and $f(a) = f(b)$. Then $f'(c) = 0$ for some c in the open interval (a, b). Ans

Proof: Since f is continuous on $[a, b]$, f attains a maximum and a minimum value on $[a, b]$. If these two extreme values are the same, then f is constant on (a, b), so $f'(x) = 0$ for all x in (a, b).

If the extreme values differ, then at least one must be attained at some point c in the open interval (a, b), and $f'(c) = 0$.

4.6 The Mean Value Theorem

Quiz

Theorem: Suppose that f is differentiable on $[a, b]$, $f'(a) \neq f'(b)$, and μ is between $f'(a)$ and $f'(b)$. Then $f'(c) = \mu$ for some c in (a, b). Ans

Proof: Suppose first that

$$f'(a) < \mu < f'(b)$$

and define

$$g(x) = f(x) - \mu x.$$

Then

$$g'(x) = f'(x) - \mu, \quad a \leq x \leq b,$$

and by our supposition $f'(a) < \mu < f'(b)$, we have

$$g'(a) < 0 \quad \text{and} \quad g'(b) > 0.$$

Since g is continuous on $[a, b]$, g attains a minimum at some point c in $[a, b]$.
Due to Lemma and $g'(a) < 0$ and $g'(b) > 0$, imply that there is a $\delta > 0$ such that

$$g(x) < g(a), \quad a < x < a + \delta, \quad \text{and} \quad g(x) < g(b), \quad b - \delta < x < b.$$

Therefore $c \neq a$ and $c \neq b$. Hence, $a < c < b$, and therefore $g'(c) = 0$, by the fact that if f is differentiable at a local extreme point x_0 then $f'(x_0) = 0$. From , $f'(c) = \mu$. The proof for the case where $f'(b) < \mu < f'(a)$ can be obtained by applying this result to $-f$.

Theorem: If f is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b), then

$$f'(c) = \frac{f(b) - f(a)}{}$$

Since $h(x) = f(g(x))$, this implies that

$$\frac{h(x) - h(x_0)}{x - x_0} = [f'(g(x_0)) + E(g(x))]\frac{g(x) - g(x_0)}{x - x_0}. \tag{4.4}$$

Since g is continuous at x_0 we have

$$\lim_{x \to x_0} E(g(x)) = E(g(x_0)) = 0.$$

Therefore, (4.4) implies that

$$h'(x_0) = \lim_{x \to x_0} \frac{h(x) - h(x_0)}{x - x_0} = f'(g(x_0))g'(x_0),$$

as stated.

Example: Calculate the derivative of

$$h(x) = \sin\frac{1}{x}, \quad x \neq 0.$$

If

$$f(x) = \sin x \quad \text{and} \quad g(x) = \frac{1}{x}, \quad x \neq 0,$$

then

$$h(x) = f(g(x)) = \sin\frac{1}{x}, \quad x \neq 0,$$

and

$$h'(x) = f'(g(x))g(x) = \left(\cos\frac{1}{x}\right)\left(-\frac{1}{x^2}\right), \quad x \neq 0.$$

Example: What is wrong in the following justification?

$$\begin{aligned}\frac{h(x) - h(x_0)}{x - x_0} &= \frac{f(g(x)) - f(g(x_0))}{x - x_0}\\ &= \frac{f(g(x)) - f(g(x_0))}{g(x) - g(x_0)}\,\frac{g(x) - g(x_0)}{x - x_0}\end{aligned}$$

and arguing that

$$\lim_{x \to x_0} \frac{f(g(x)) - f(g(x_0))}{g(x) - g(x_0)} = f'(g(x_0))$$

(because $\lim_{x \to x_0} g(x) = g(x_0)$) and

$$\lim_{x \to x_0} \frac{g(x) - g(x_0)}{x - x_0} = g'(x_0).$$

Is it a valid proof?

if the limit exists.

The definition of limit implies that f is differentiable at x_0 if and only if $f'_+(x_0)$ and $f'_-(x_0)$ exist and are equal, in which case

$$f'(x_0) = f'_+(x_0) = f'_-(x_0).$$

The functions defined by $f(x) = |x|$ and $f(x) = \frac{|x|}{x}$ have one sided derivatives.

Example: For the piecewise defined function

$$f(x) = \begin{cases} x^3, & x \le 0, \\ x^2 \sin \frac{1}{x}, & x > 0, \end{cases}$$ Quiz

Investigate the one sided derivatives.

Solution: For the given function, we have

$$f'(x) = \begin{cases} 3x^2, & x < 0, \\ 2x \sin \frac{1}{x} - \cos \frac{1}{x}, & x > 0. \end{cases}$$

Since neither formula in $f(x)$ holds for all x in any neighborhood of 0, we cannot simply differentiate either to obtain $f'(0)$.

We will calculate the one sided derivatives

$$\begin{aligned} \lim_{x \to 0+} f(x) &= \lim_{x \to 0+} \frac{x^2 \sin \frac{1}{x} - 0}{x - 0} = \lim_{x \to 0+} x \sin \frac{1}{x} = 0, \\ f'_-(0) &= \lim_{x \to 0-} \frac{x^3 - 0}{x - 0} = \lim_{x \to 0-} x^2 = 0; \end{aligned}$$

hence, $f'(0) = f'_+(0) = f'_-(0) = 0.$

Remark: There is a difference between one sided derivatives and one sided limit of derivative

$$f'' = f'', \quad f^{(3)} = f''', \quad \frac{d^n f}{dx^n} = f^{(n)}$$

Example: If n is a positive integer, find the derivative of the function

$$f(x) = x^n.$$

Solution: We have

$$\frac{f(x) - f(x_0)}{x - x_0} = \frac{x^n - x_0^n}{x - x_0} = \frac{x - x_0}{x - x_0} \sum_{k=0}^{n-1} x^{n-k-1} x_0^k.$$

$$f'(x_0) = \lim_{x \to x_0} \sum_{k=0}^{n-1} x^{n-k-1} x_0^k = n x_0^{n-1}.$$

Since this holds for every x_0, we drop the subscript and write

$$f'(x) = n x^{n-1} \quad \text{or} \quad \frac{d}{dx}(x^n) = n x^{n-1}.$$ Ans

Sol

Example: Find the derivative of the line $y = mx + c$.

Solution:

$$\begin{aligned} &\frac{f(x) - f(x_0)}{x - x_0} \\ &= \frac{mx + c - (mx_0 + c)}{x - x_0} \\ &= \frac{m(x - x_0)}{x - x_0} = m \end{aligned}$$

Consequently

$$f'(x_0) = \lim_{x \to x_0} m = m.$$

Geometrical interpretation of derivative: The equation of the line through two points $(x_0, f(x_0))$ and $(x_1, f(x_1))$ on the curve $y = f(x)$ is

$$y = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0).$$

fore only one value of x_0 can satisfy $f(x_0) = y_0$, for each y_0.

To see that g is increasing, suppose that $y_1 < y_2$ and let x_1 and x_2 be the points in $[a, b]$ such that $f(x_1) = y_1$ and $f(x_2) = y_2$.

Since f is increasing, $x_1 < x_2$. Therefore,

$$g(y_1) = x_1 < x_2 = g(y_2),$$

so g is increasing.

Since $R_g = \{g(y) : y \in [c, d]\}$ is the interval $[g(c), g(d)] = [a, b]$, therefore from theorem, we have proved with f and $[a, b]$ replaced by g and $[c, d]$ implies that g is continuous on $[c, d]$.

Example: If

Quiz

$$f(x) = x^2, \quad 0 \le x \le R.$$

Solution: The inverse of the given function is

$$f^{-1}(y) = g(y) = \sqrt{y}, \quad 0 \le y \le R^2.$$

Ans

Example: If

$$f(x) = 2x + 4, \quad 0 \le x \le 2,$$

Solution: The inverse of the given function is

Quiz

$$f^{-1}(y) = g(y) = \frac{y-4}{2}, \quad 4 \le y \le 8.$$